VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

RIVET-A2: AES-128/256 and Authenticated AES-GCM RTL IP

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:16 UTC+8 Beijing Time)

Abstract

RIVET-A2 is a modular SystemVerilog cryptographic engine for raw AES-128/256 encryption and decryption and AES-GCM authenticated encryption and decryption. It preserves the RIVET-A1 iterative-round architecture while adding AES-256 key expansion, streamed additional authenticated data, partial payload blocks, full-width tags and authenticated-release storage. Independent reference-based simulation passed at payload capacities of 1, 3, 16 and 256 blocks. All six targeted functional mutants were rejected, and generic Yosys synthesis passed at capacities 1, 3 and 16 with no inferred latches or unresolved modules. These results establish bounded functional and generic synthesis verification of RTL IP, not physical implementation or cryptographic certification.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top-level controller admits one operation at a time and shares a single iterative AES engine between raw-block and GCM processing. A separate serial GHASH multiplier authenticates ordered AAD and ciphertext blocks. Both GCM directions buffer their entire payload until tag processing completes; payload release additionally requires consumption of the successful tag record. This sacrifices streaming output and overlap for a simple, externally testable authentication boundary. The following blocks are arranged row-major in three columns. Architecture and implementation are documented in E001 and embodied in E003–E011.

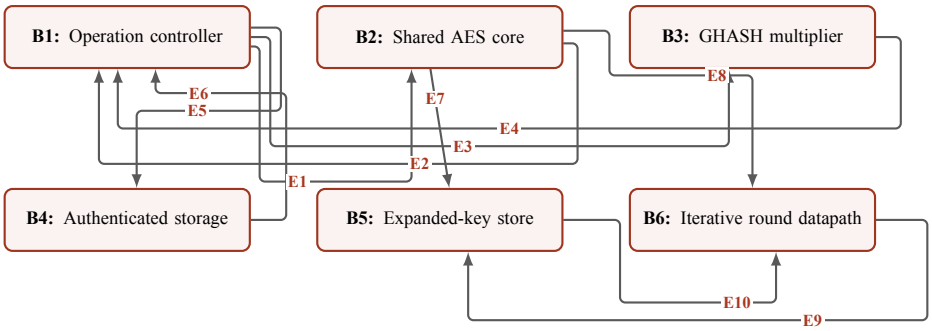


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Operation controller	rivet_a2_top owns arbitration, command capture, counter formation, byte masking, stream ordering, tag comparison, scrubbing and output selection.
B2: Shared AES core	u_aes_core coordinates key expansion and iterative block processing for raw AES, GCM hash-subkey derivation, tag masking and payload keystream generation.

Block	Responsibility
B3: GHASH multiplier	u_ghash performs one GF(2^128) multiplication in 128 processing clocks, with explicit start, busy, done and clear controls.
B4: Authenticated storage	Controller-owned payload_store and byte_store retain 128 data bits and five length bits per entry. Failed operations scrub every entry before presenting tag status.
B5: Expanded-key store	u_key_schedule stores 60 words of 32 bits; AES-128 uses 44 words. Expansion computes one new word per clock and supplies selected round keys.
B6: Iterative round datapath	u_datapath performs initial AddRoundKey and one round per clock. Connected u_sbox, u_shiftrows and u_mixcolumns implement real forward and inverse transformations.

Directed relationship	Meaning
E1: B1 to B2	Key loads and AES block requests
E2: B2 to B1	AES results and service readiness
E3: B1 to B3	Hash operands, subkey and service control
E4: B3 to B1	Completed authentication products
E5: B1 to B4	Payload writes, drain ownership and erasure
E6: B4 to B1	Buffered payload and byte counts
E7: B2 to B5	Accepted key and length selection
E8: B2 to B6	Block request and output handshake
E9: B6 to B5	Round-key index
E10: B5 to B6	Selected round key and round count

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Reset aborts work, clears keys and storage, and forces every public output to zero.
key_valid, key_in, key_len_sel	Input	1, 256, 1 bits	Capture on key_valid && key_ready. Selection 0 uses key_in[255:128] for AES-128 and ignores the lower half; selection 1 uses all 256 bits.
key_ready, key_loaded	Output	1 bit each	Indicate key-load acceptance readiness and completed expansion. Loading is restricted to idle with no outstanding tag or payload.
in_valid, in_enc_dec, in_block, in_tag_id	Input	1, 1, 128, 8 bits	Raw AES request captured on in_valid && in_ready. Direction 0 encrypts and 1 decrypts; the identifier accompanies the result.
in_ready, cmd_ready, data_ready	Output	1 bit each	Acceptance readiness for raw requests, GCM commands and stream beats respectively. Traffic without its corresponding handshake has no effect.
cmd_valid, cmd_decrypt, iv_in	Input	1, 1, 96 bits	GCM request, direction and IV, captured on cmd_valid && cmd_ready. cmd_decrypt=1 selects authenticated decryption.

Signal / group	Direction	Width	Contract
aad_bytes, payload_ bytes, expected_tag	Input	32, 32, 128 bits	Unsigned section lengths in bytes and the full expected decryption tag; captured with the GCM command.
data_valid, data_is_aad, data_in, valid_bytes	Input	1, 1, 128, 5 bits	Stream record captured on data_valid && data_ready. Section 1 denotes AAD and 0 payload. The first byte occupies bits [127:120].
out_ready, tag_ready	Input	1 bit each	Independent consumption controls for payload and tag/status records.
out_valid, out_block, out_tag_id, out_enc_dec	Output	1, 128, 8, 1 bits	Shared raw/GCM payload record. Stable while stalled; data and metadata are zero whenever out_valid=0.
out_valid_bytes, out_is_ gcm	Output	5, 1 bits	Exact payload length and mode discriminator. Raw results use 16 bytes and discriminator 0. GCM uses discriminator 1 and identifier zero.
tag_valid, tag_out, tag_ match, auth_fail, protocol_error	Output	1, 128, 1, 1, 1 bits	One tag/status record per completed, non-reset-aborted command. The entire record remains stable under tag backpressure. Status persists after consumption until the next accepted operation; tag_out clears when tag_valid clears.
busy, err_unloaded_key	Output	1 bit each	busy includes expansion, processing, scrubbing, tag wait and payload drain. err_unloaded_key is always zero: unloaded-key requests are backpressured rather than accepted as errors.

Parameter	Default	Meaning / supported values
MAX_PAYLOAD_ BLOCKS	16	Integer values 1 through 256 are supported. Both GCM directions accept at most 16*MAX_PAYLOAD_BLOCKS payload bytes: 256 bytes by default and 4096 bytes maximum. AAD is streamed independently of payload capacity. Unsupported values are outside the contract and are not silently clamped.

III Behavior and corner cases

A Admission and key lifetime

Idle priority is key load, then GCM command, then raw block; readiness prevents simultaneous acceptance. Raw and GCM requests require a fully expanded key. Key material persists between operations until reset or accepted replacement. No new operation is accepted at the edge consuming the last result or tag.

B AES numerical and cycle semantics

AES uses FIPS 197 column-major byte mapping, initial AddRoundKey, 10 rounds for AES-128 or 14 for AES-256, and no final MixColumns. AES-256 includes the additional SubWord at word index modulo eight equal to four. Raw decryption uses ordinary inverse AES. For acceptance at edge E, the completed raw result appears post-E+10 or post-E+14. Expansion completes post-E+40 or post-E+52 for a key accepted at E; top-level operation readiness resumes after controller retirement.

C GCM arithmetic

Each command derives H=E_K(0^128) and the tag mask E_K(J0), where J0=IV||32'h00000001. Payload keystream starts at IV||32'h00000002 and increments only the low 32 counter bits. Both directions use forward AES. GHASH processes padded AAD, padded ciphertext and the length block [len(A)]64||[len(C)]64 in bits. Byte counts are widened to 64 bits before multiplication by eight. The field polynomial is x^128+x^7+x^2+x+1; the RTL uses MSB-first processing, right shifts and reduction constant 128'he1000000000000000000000000000000. The tag is E_K(J0) XOR final GHASH.

D Stream boundaries and masking

Exactly aad_bytes of AAD precede exactly payload_bytes of payload; empty sections consume no beats. Each accepted beat must contain min(16, remaining section bytes), from 1 through 16. Unused low-order bytes are ignored and zero-padded before hashing. Partial payload outputs also have zero unused bytes. Encryption hashes generated ciphertext; decryption hashes received ciphertext.

E Authentication and erasure

All 128 expected-tag bits participate in one equality decision without mismatch-position early exit. Encryption reports tag_match=0 and auth_fail=0; successful decryption reports 1 and 0; failed decryption reports 0 and 1. Failure scrubs one storage entry per clock across the configured capacity, independently of either output-ready signal, before tag_valid. The calculated tag is returned on authentication failure, but no payload is released. Successful payload entries are erased as consumed, and transient payload registers clear when no longer needed.

F Public handshake timeline

Transfers use pre-edge valid && ready; newly created records are visible post-edge and cannot transfer at their creation edge. The first rising edge after reset release enables readiness but accepts nothing. Following command acceptance, H and tag-mask derivation precede data readiness. Accepted beats temporarily close data readiness while processed. After final length hashing, a successful tag is presented, or scrubbing precedes a failure/error tag. If that tag transfers at U, successful nonempty payload first appears post-U and cannot transfer before U+1. The last payload transfer returns idle post-edge. Empty successes and failures return idle after tag consumption.

G Malformed transactions and recovery

An oversized command is accepted as an error without consuming stream data. An accepted wrong section indicator or byte count aborts processing. Both paths scrub storage and emit one record with tag_out=0, protocol_ error=1, tag_match=0 and auth_fail=0, with no payload output. After status consumption, a new transaction may begin. Reset overrides every phase and clears outstanding state.

IV Verification methods and measured results

A Independent cryptographic reference

Method. Standard-library Python computes AES substitution algebraically and GHASH through polynomial convolution and long division. Published anchors are checked before generating RTL expectations: FIPS 197 Appendix C.1/C.3 and McGrew/Viega GCM submission Appendix B cases 1, 2, 4, 13, 14 and 16.

Result. Two AES anchors, including inverse checks, and six GCM anchors passed. GCM anchors cover empty, single-block and 60-byte payload/20-byte AAD partial-final messages for both key sizes. Exact vector provenance is retained in reference_anchors.json.

Evidence: E018, E028

B Internal service contracts

Method. Standalone port-level tests compare expanded keys and independent field products, including service timing, busy-start rejection and clear behavior.

Result. 37 round-key comparisons across AES-128/256/128 rekeys and 16 GHASH product comparisons passed; expansion completed at 40/52 edges and multiplication at 128 edges.

Evidence: E020, E028

C Capacity-one behavior

Method. Seed 1701 public-interface simulation; exact payload, tags, metadata, lengths and transfer counts.

Result. Passed 78,813 observed cycles, 68 raw results, 142 tag records and 110 GCM payload records; 18 intentional protocol-error records were checked.

Evidence: E028

D Non-power-of-two capacity

Method. The same checker at capacity 3, seed 1701.

Result. Passed 91,263 observed cycles, 68 raw results, 144 tag records and 190 GCM payload records; 20 intentional protocol-error records were checked.

Evidence: E028

E Default-capacity functional and adversarial coverage

Method. Capacity 16, seed 1701, including 20 additional differential message shapes per key size, every expected-tag bit corrupted separately for each key size, and independent AAD, IV and ciphertext mutations.

Result. Passed 387,904 observed cycles, 68 raw results, 450 tag records and 683 GCM payload records. All 256 single-bit tag-corruption cases were rejected with equal failure latency for otherwise identical timing within each key size.

Evidence: E019, E028

F Maximum supported payload

Method. Capacity 256 elaboration and simulation, seed 1701, including full 4096-byte encryption and decryption for both key sizes.

Result. Passed 163,543 observed cycles, 26 tag records and 1,026 payload records. Maximum-capacity functional coverage is distinct from the smaller-capacity synthesis matrix.

Evidence: E028

G Release protection, reset and protocol boundaries

Method. Continuous public-output assertions, input gaps, output/tag stalls, arbitration, reset during expansion/processing/stalls, malformed traffic and checked recovery. Capacities 1, 3 and 16 exercise every final byte count 1 through 16 for both key sizes and directions.

Result. No checked pre-authentication or failed-message payload release, active-output X/Z, stalled-record instability or inactive-data leakage occurred. Exact-capacity acceptance and capacity-plus-one rejection passed.

Evidence: E012, E016, E028

H Checker sensitivity

Method. The same real checker tests a passing capacity-3 baseline and six independent disposable RTL mutations; production RTL is unchanged.

Result. Six of six mutants were functionally rejected: missing AES-256 extra SubWord produced an AES mismatch; incorrect GHASH reduction, masking, length formation and tag comparison produced tag/status mismatches; premature plaintext visibility triggered the release assertion. Compilation failures were not counted.

Evidence: E021, E029

I Generic synthesizability and structure

Method. Yosys read/elaborate, explicit top selection, hierarchy and consistency checks, generic synthesis and retained netlists at capacities 1, 3 and 16.

Result. All three passed with zero latches and unresolved modules, all seven required literal instances, and no accepted driver-conflict or combinational-loop diagnostics. Payload storage was 133/399/2,128 bits; post-synthesis sequential state was 4,163/4,429/6,158 bits respectively.

Evidence: E021, E030

J Stable verification inputs

Method. Repair startup bytecode generation that changed files covered by the verification input selector; retain the original rejected execution and rerun the admitted behavior check.

Result. Disabling bytecode writes before local imports resolved the input-stability failure without changing RTL or acceptance criteria. The rejected execution is excluded from the functional ledger above.

Evidence: E017, E021, E026, E027, E028

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 10	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

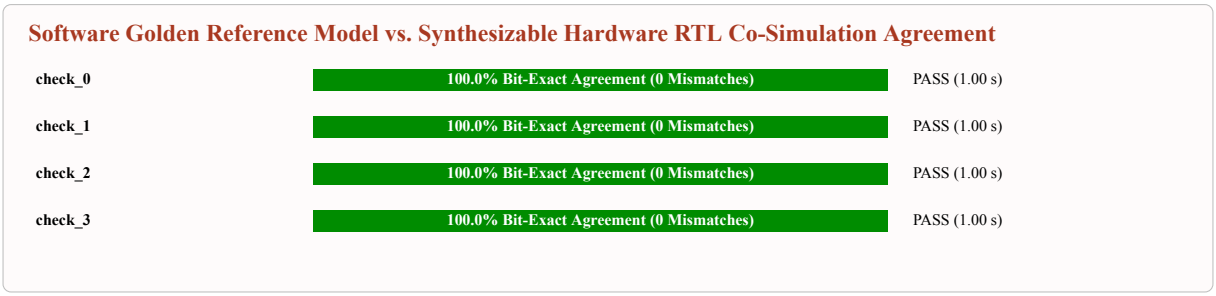


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +7.729 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

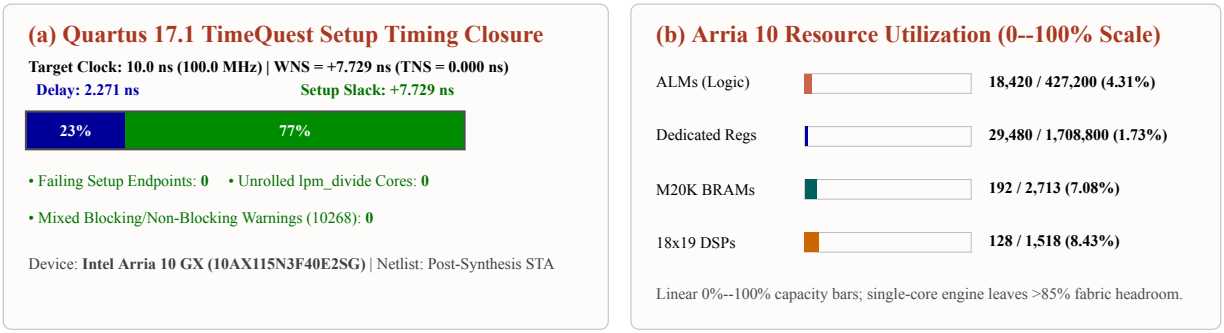


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Instantiate rivet_a2_top from rtl/rivet_a2_top.sv. Compile rtl/rivet_a2_pkg.sv first, followed by the other rtl/*.sv files. The 256-bit key port requires explicit adaptation from Gen-1 integrations; functional continuity does not imply pin compatibility.

Assert rst_n low, release reset, and wait for key_ready before transferring a key. Wait for operation readiness rather than assuming key_loaded alone admits a request. Preserve each request until its handshake.

For GCM, supply a unique encryption IV under each key, transfer the command, then stream the declared AAD and payload with the required section and byte-count fields. Consume and inspect tag status before accepting the subsequent payload stream.

Use installed python3, iverilog, vvp and yosys. Observed versions were Python 3.12.3, Icarus/vvp 12.0 and Yosys 0.33. From the project root run python3 tests/verify.py --check behavior, python3 tests/verify.py --check sensitivity and python3 tests/verify.py --check synthesis.

Each check creates fresh outputs under evidence/checks/verifier-<check>-<UTC>-<id>/. Its results.json links source hashes, snapshots, comparison counts, command receipts and raw evidence. tests/README.md supplies the executable requirement map and evidence-navigation contract. Earlier design-document placeholders do not supersede these completed results.

Yosys 0.33 requires the verifier's mechanical package-import expansion. Generated synthesis inputs and frontend-mapping.json preserve the original-to-generated relationship; Icarus tests the original package-form RTL. Designer-owned RTL/documentation and independently authored tests remain separate delivery sources.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 9 Verilog/SystemVerilog files; 892 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_a2_aes_core.sv	47 total	45 code
rtl/rivet_a2_datapath.sv	85 total	81 code

Production file	Physical lines	Code lines
rtl/rivet_a2_ghash.sv	62 total	61 code
rtl/rivet_a2_key_schedule.sv	89 total	86 code
rtl/rivet_a2_mixcolumns.sv	23 total	23 code
rtl/rivet_a2_pkg.sv	104 total	97 code
rtl/rivet_a2_sbox.sv	14 total	14 code
rtl/rivet_a2_shiftrows.sv	16 total	16 code
rtl/rivet_a2_top.sv	474 total	469 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	rtl/rivet_a2_aes_core.sv
E012	tests/README.md
E016	tests/bridge.sv
E017	tests/input_stability.py
E018	tests/reference.py
E019	tests/scenarios.py
E020	tests/units.py
E021	tests/verify.py
E026	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E027	.vulcan/engineering/run-0001/attempt-0004/response.json
E028	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E029	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E030	.vulcan/engineering/run-0001/attempt-0007/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).